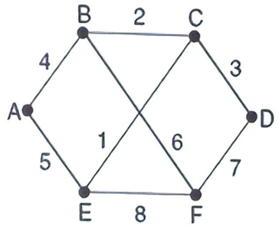
5. Consider the network of Fig. 5-12(a). Distance vector routing is used, and the following vectors have just come in to router C: from B: (5, 0, 8, 12, 6, 2); from D: (16, 12, 6, 0, 9, 10); and from E: (7, 6, 3, 9, 0, 4). The cost of the links from C to B, D, and E, are 6, 3, and 5, respectively. What is C's new routing table? Give both the outgoing line to use and the cost.




Solution for Problem 5-5

| router | C |
|---|---|

| to | | A | B | C | D | E | F | cost |
|---|---|---|---|---|---|---|---|---|
| received routing table from | B | 5 | 0 | 8 | 12 | 6 | 2 | 6 |
| | D | 16 | 12 | 6 | 0 | 9 | 10 | 3 |
| | E | 7 | 6 | 3 | 9 | 0 | 4 | 5 |

(5, 0, 8, 12, 6, 2)
(16, 12, 6, 0, 9, 10)
(7, 6, 3, 9, 0, 4)

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| new cost from | B | 11 | 6 | 14 | 18 | 12 | 8 |
| | D | 19 | 15 | 9 | 3 | 12 | 13 |
| | E | 12 | 11 | 8 | 14 | 5 | 9 |

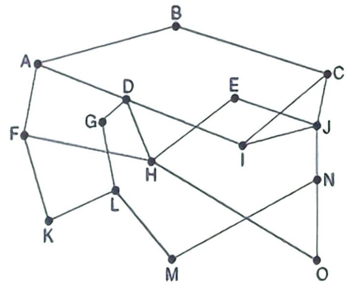
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| minimum cost | | 11 | 6 | 0 | 3 | 5 | 8 |
| outgoing line | | B | B | - | D | E | B |

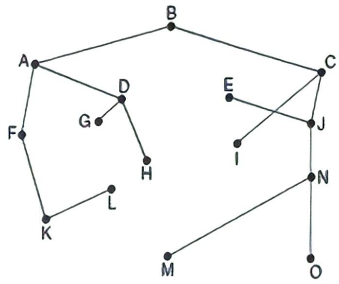
New Routing Table for router C
cost: (11, 6, 0, 3, 5, 8)
line: (B, B, -, D, E, B)

9. Looking at the network of Fig. 5-6, how many packets are generated by a broadcast from B, using

(a) reverse path forwarding?

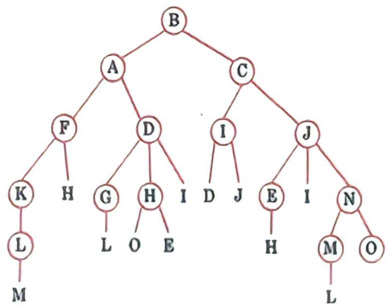
(b) the sink tree?



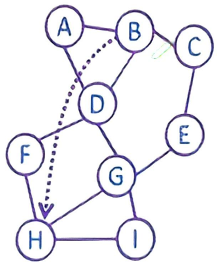




(a) The reverse path forwarding algorithm: 5 rounds and 25 packets.

(b) The sink tree: needs 4 rounds and 14 packets.



12. Suppose that node B in Fig. 5-20 has just rebooted and has no routing information in its tables. It suddenly needs a route to H. It sends out broadcasts with TTL set to 1, 2, 3, and so on. How many rounds does it take to find a route?




Node H is three hops from B, so it takes three rounds to find the route.

15. Describe two major differences between the ECN method and the RED method of congestion avoidance.

The main differences between ECN and RED are:

- Strategy for **congestion notification**
  - ECN, explicit notification;
  - RED, implicit notification;
- Strategy for **dropping packet**
  - ECN, never discard before cache overflow;
  - RED, randomly discarded before cache overflow;
- Strategy for **source quench**
  - ECN, when receiving the ECE, it begins to slow down;
  - RED, after the packet is lost, it begins to slow down;

| | **ECN** | **RED** |
|---|---|---|
| Congestion Notification | Explicit | Implicit |
| Dropping Packet | never discard before cache overflow | randomly discarded before cache overflow |
| Source Quench | when receiving the ECE | after the packet is lost |

23. Suppose that instead of using 16 bits for the network part of a class B address originally, 20 bits had been used. How many class B networks would there have been?

With a 20-bit prefix

$$2^{20-2} = 2^{18} = 262{,}144\ (network)$$

all 0s (this network) and all 1s (all hosts) are special

$$262{,}144 - 2 = 262{,}142\ (network)$$

24. Convert the IP address whose hexadecimal representation is C22F1582 to dotted decimal notation.

C2.2F.15.82

$= (12 \times 16 + 2).(2 \times 16 + 15).(1 \times 16 + 5).(8 \times 16 + 2)$

$= 194.47.21.130$

25. A network on the Internet has a subnet mask of 255.255.240.0. What is the maximum number of hosts it can handle?

$$240_{DEC} = 1111\ 0000_{BIN}$$

$$2^{4+8} = 4{,}096$$

all 0s (network id) and all 1s (broadcast) are special

$$4{,}096 - 2 = 4{,}094$$

27. A large number of consecutive IP addresses are available starting at 198.16.0.0. Suppose that four organizations, A, B, C, and D, request 4000, 2000, 4000, and 8000 addresses, respectively, and in that order. For each of these, give the first IP address assigned, the last IP address assigned, and the mask in the w.x.y.z/s notation.

$2^{10} = 1024,\quad 2^{11} = 2048,\quad 2^{12} = 4096,\quad 2^{13} = 8192$

```
11000110 00010000 00000000 00000000
11000110 00010000 00001111 11111111 , 4000 addrs for A
11000110 00010000 00010000 00000000
11000110 00010000 00010111 11111111 , 2000 addrs for B
11000110 00010000 00011000 00000000
11000110 00010000 00011111 11111111 , 2048 addrs reserved
11000110 00010000 00100000 00000000
11000110 00010000 00101111 11111111 , 4000 addrs for C
11000110 00010000 00110000 00000000
11000110 00010000 00111111 11111111 , 4096 addrs reserved
11000110 00010000 01000000 00000000
11000110 00010000 01011111 11111111 , 8000 addrs for D
```

A: 198.16.0.0 – 198.16.15.255 written as 198.16.0.0/20 , 4000 addresses

B: 198.16.16.0 – 198.16.23.255 written as 198.16.16.0/21 , 2000 addresses

C: 198.16.32.0 – 198.16.47.255 written as 198.16.32.0/20 , 4000 addresses

D: 198.16.64.0 – 198.16.95.255 written as 198.16.64.0/19 , 8000 addresses

28. A router has just received the following new IP addresses: 57.6.96.0/21, 57.6.104.0/21, 57.6.112.0/21, and 57.6.120.0/21. If all of them use the same outgoing line, can they be aggregated? If so, to what? If not, why not?

| | |
|---|---|
| 57.6.96.0/21, | 0011 1001 0000 0110 0110 0000 0000 0000 |
| 57.6.104.0/21, | 0011 1001 0000 0110 0110 1000 0000 0000 |
| 57.6.112.0/21, | 0011 1001 0000 0110 0111 0000 0000 0000 |
| 57.6.120.0/21 | 0011 1001 0000 0110 0111 1000 0000 0000 |

They can be aggregated to 57.6.96.0/19.

29. The set of IP addresses from 29.18.0.0 to 29.18.127.255 has been aggregated to 29.18.0.0/17. However, there is a gap of 1024 unassigned addresses from 29.18.60.0 to 29.18.63.255 that are now suddenly assigned to a host using a different outgoing line. Is it now necessary to split up the aggregate address into its constituent blocks, add the new block to the table, and then see if any reaggregation is possible? If not, what can be done instead?

$60_{DEC} = 0011\ 1100_{BIN}$

$64_{DEC} = 0100\ 0000_{BIN}$

It is sufficient to add one new table entry: 29.18.60.0/22 for the new block.

If an incoming packet matches both 29.18.0.0/17 and 29.18.60.0/22, the longest one wins.

30. A router has the following (CIDR) entries in its routing table:

| Address/mask | Next hop | |
|---|---|---|
| 135.46.56.0/22 | Interface 0 | $56_{DEC} = 0011\ 1000_{BIN}$ |
| 135.46.60.0/22 | Interface 1 | $60_{DEC} = 0011\ 1100_{BIN}$ |
| 192.53.40.0/23 | Router 1 | $40_{DEC} = 0010\ 1000_{BIN}$ |
| Default | Router 2 | |

For each of the following IP addresses, what does the router do if a packet with that address arrives?

(a) 135.46.63.10 → Interface 1
(b) 135.46.57.14 → Interface 0
(c) 135.46.52.2 → Router 2
(d) 192.53.40.7 → Router 1
(e) 192.53.56.7 → Router 2

35. In IP, the checksum covers only the header and not the data. Why do you suppose this design was chosen?

Ensuring the correctness of the header is the responsibility of the network layer.

- Errors in the header may lead to unpredictable consequences. The router is heavily burdened and unable to check for errors in the data.
- Errors in the data can be checked by the upper level.

37. IPv6 uses 16-byte addresses. If a block of 1 million addresses is allocated every picosecond, how long will the addresses last?

$2^{16\times8} = 3.4\times10^{38}$

$\frac{3.4\times10^{38}}{10^{6}\times10^{12}} = 3.4\times10^{20}\ (sec) = 1.08\times10^{13}\ (year)$

38. The Protocol field used in the IPv4 header is not present in the fixed IPv6 header. Why not?

Protocol field in IPv4 header

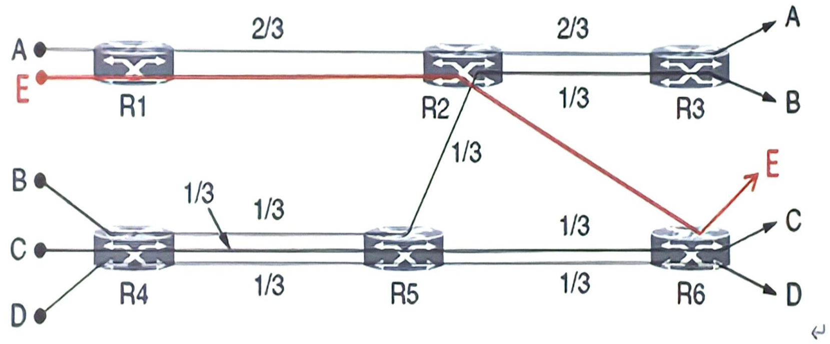
- The router does not process this field, it will be processed on the destination host.

The next header field is used in the fixed header of IPv6.

- The router reads this field for extension of basic header, it will be processed on the destination host.

7.

8. In Figure 6-20, suppose a new flow E is added that takes a path from R1 to R2 to R6. How does the max-min bandwidth allocation change for the five flows?

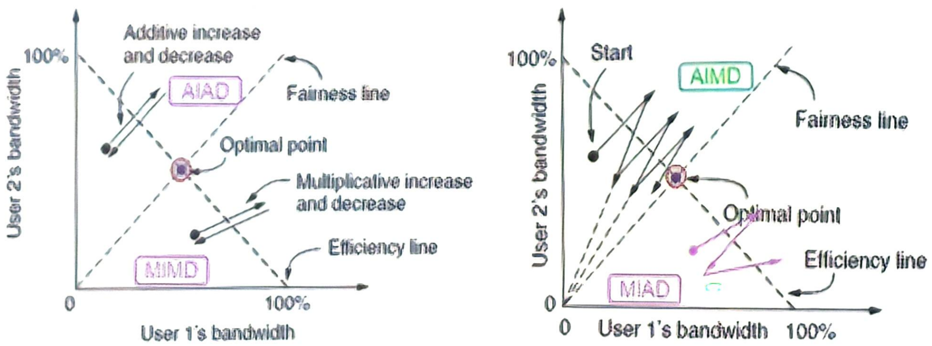



Allocation for flow A: 1/2, on links R1 → R2 → R3.

Allocation for flow E: 1/2, on links R1 → R2 → R6.

All other allocations remain the same.

10. Some other policies for fairness in congestion control are Additive Increase Additive Decrease (AIAD), Multiplicative Increase Additive Decrease (MIAD), and Multiplicative Increase Multiplicative Decrease (MIMD). Discuss these three policies in terms of convergence and stability.




AIAD: oscillate, not converge, and not be stable.

MIMD: oscillate, not converge, and not be stable.

AIMD: converge, and be stable.

MIAD: converge, but not be stable.

11. Why does UDP exist? Would it not have been enough to just let user processes send raw IP packets?

No.

IP packets contain IP addresses, which specify a destination machine.
UDP packets contain a destination port. It can be delivered to the correct process.

15. Both UDP and TCP use port numbers to identify the destination entity when delivering a message. Give two reasons why these protocols invented a new abstract ID (port numbers), instead of using process IDs, which already existed when these protocols were designed.

First, process IDs are OS-specific. Using process IDs would have made these protocols OS-dependent.

Second, a single process may establish multiple channels of communications. A single process ID (per process) as the destination identifier cannot be used to distinguish between these channels.

Third, having processes listen on well-known ports is easy, but well-known process IDs are impossible.

18. What is the total size of the minimum TCP MTU, including TCP and IP overhead but not including data link layer overhead?

The default segment is 536 bytes.
TCP adds 20 bytes
IP adds 20 bytes
the total size of min TCP MTU is 576 bytes.

19. Datagram fragmentation and reassembly are handled by IP and are invisible to TCP. Does this mean that TCP does not have to worry about data arriving in the wrong order?

No, this doesn't mean that. or

Yes, TCP has to worry about it.

IP reassembly can only solve the problem of a single packet;

TCP aims to solve the problem of the entire message, so it must be concerned about whether IP packets arrive in order.

21. A process on host 1 has been assigned port p, and a process on host 2 has been assigned port q. Is it possible for there to be two or more TCP connections between these two ports at the same time?

No.

A connection is identified only by its sockets. {IP1, Port1} ↔ {IP2, Port2}

Thus, (1, p) – (2, q) is the only possible connection between those two ports.

22. In Fig. 6-36 we saw that in addition to the 32-bit acknowledgement field, there is an ACK bit in the fourth word. Does this really add anything? Why or why not?

The ACK bit is used to tell whether the 32-bit acknowledgement field is used.

It is not absolutely essential for normal data traffic.

However, it plays a crucial role during connection establishment, where it is used in the second and third messages of the three-way handshake.

25. Consider the effect of using slow start on a line with a 10-msec round-trip time and no congestion. The receive window is 24 KB and the maximum segment size is 2 KB. How long does it take before the first full window can be sent?

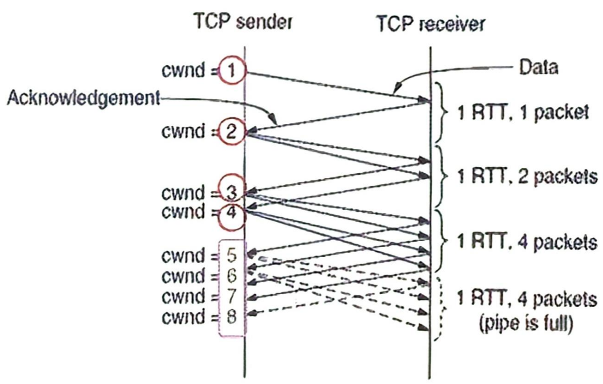
1) 0, 2KB
2) 10ms, 4KB
3) 20ms, 8KB
4) 30ms, 16KB

5) 40ms, 24KB (32KB)

26. Suppose that the TCP congestion window is set to 18 KB and a timeout occurs. How big will the window be if the next four transmission bursts are all successful? Assume that the maximum segment size is 1 KB.

1) 1KB
2) 2KB
3) 4KB
4) 8KB

5) 9KB (18KB/2)



27. If the TCP round-trip time, RTT, is currently 30 msec and the following acknowledgements come in after 26, 32, and 24 msec, respectively, what is the new RTT estimate using the Jacobson algorithm? Use $\alpha = 0.9$.

$$SRTT = \alpha \cdot SRTT + (1 - \alpha) \cdot R$$

1st round: $0.9 \times 30\ (ms) + (1 - 0.9) \times 26\ (ms) = 29.6\ (ms)$

2nd round: $0.9 \times 29.6\ (ms) + (1 - 0.9) \times 32\ (ms) = 29.84\ (ms)$

3rd round: $0.9 \times 29.84\ (ms) + (1 - 0.9) \times 24\ (ms) = 29.256\ (ms)$

32. A CPU executes instructions at the rate of 1000 MIPS. Data can be copied 64 bits at a time, with each word copied costing 10 instructions. If an incoming packet has to be copied four times, can this system handle a 1-Gbps line?

For simplicity, assume that all instructions, even those instructions that read or write memory, run at the full 1000-MIPS rate.

System capacity: $64\ (bit) \times 1000 \times 10^6\ /\ (10 \times 4) = 1.6 \times 10^9 (bps)$

It can handle a 1-Gbps line.

33. To get around the problem of sequence numbers wrapping around while old packets still exist, one could use 64-bit sequence numbers. However, theoretically, an optical fiber can run at 75 Tbps.

What maximum packet lifetime is required to make sure that future 75-Tbps networks do not have wrap around problems even with 64-bit sequence numbers? Assume that each byte has its own sequence number, as TCP does.

$$\frac{2^{64}}{75 \times 10^{12}\ /\ 8} = 1.968 \times 10^6 (s) = 22.77 (day)$$

S-1. Suppose a link capacity is 1000Mbps, there are four streams A, B, C, and D, which require traffic capacity of 100Mbps, 200Mbps, 400Mbps, and 500Mbps respectively. What are the capacities of each stream under the max-min fair allocation?

1st Round: 100, 100, 100, 100 (Mbps)

2nd Round: 100, 200, 200, 200 (Mbps)

3rd Round: 100, 200, 350, 350 (Mbps)

Textbook, P.777 Problems for Chapter 7

1. In Fig. 7-4, there is no period after laserjet. Why not?

It is not an absolute name, but relative to .cs.vu.nl.

It is really just a shorthand notation for laserjet.cs.vu.nl.

2. Consider a situation in which a cyberterrorist makes all the DNS servers in the world crash simultaneously. How does this change one's ability to use the Internet?

The DNS servers provide a mapping between domain names and IP addresses.

If all the DNS servers in the world were to crash at the same time, one would not be able to map between domain names and IP addresses.

Therefore, the only way to access Web pages would be by using the IP address of the host server instead of the domain name.

Since most of us do not know the IP addresses of the servers we access, this type of situation would make use of the Internet extremely inefficient.